

The Participation Puzzle: Gender Disparities in Unpaid Care Work in India

Dr Preet Deep Singh

2025-10-17

Abstract

We decompose gender disparities in unpaid care work using India's 2024 Time Use Survey (10.2 million observations, 139,489 households). Women perform unpaid care work at rates 3.5 times higher than men, yet conditional on participation, men and women work similar durations. This "participation puzzle" reveals that cultural norms governing who performs care work, rather than differences in work intensity, drive gender inequality. The pattern persists across education, employment, and household composition. Regression analysis confirms female gender predicts participation independent of all controls. Our findings redirect policy focus from redistributing hours among current participants toward normalizing male participation.

JEL Classification: J16, J22, D13, O15

Keywords: unpaid work, gender inequality, time use, care work, India

1 Introduction

Gender inequality in unpaid care work constitutes a fundamental mechanism reproducing economic disadvantage across generations. Women worldwide devote substantially more time to unpaid domestic and care activities than men, constraining their labor force participation,

earnings potential, and career advancement (Ferrant et al. 2014; Folbre 2006). Yet the precise nature of this disparity—whether rooted in differential participation rates or differential time intensity conditional on participation—remains empirically underexplored.

This distinction matters. If gender gaps stem from women working longer hours when performing care activities, interventions might focus on redistributing workload among current participants through household bargaining or service provision. If gaps instead reflect lower male participation rates at similar intensities, policy must target the cultural norms and institutional structures that exempt men from care responsibilities entirely.

We exploit India’s 2024 Time Use Survey—a nationally representative dataset capturing 10.2 million activity observations across 139,489 households—to decompose gender disparities in unpaid care work along participation and intensity margins. Our analysis yields a striking pattern: women perform unpaid care work at rates 3.5 times higher than men (4.78% versus 1.56% daily participation), yet conditional on participation, men and women devote statistically indistinguishable time to care activities (50.7 versus 47.4 minutes per activity). We term this the “participation puzzle.”

1.1 Contributions

We extend this core result in several directions. First, we demonstrate that the participation gap persists across urban-rural divides, educational strata, employment status, and household structures, suggesting deep cultural rather than purely economic determinants. Second, we document substantial heterogeneity in the participation gap by presence of young children and elderly household members—contexts where care demands are highest. Third, regression analysis confirms that female gender predicts participation independent of age, marriage, education, employment, and household composition, with an effect size exceeding 30 percentage points. Fourth, our findings inform the gender economics literature by identifying participation barriers as the binding constraint on equality in care work provision

(Goldin 2014; Bertrand 2011).

2 Literature Review

2.1 Gender and Unpaid Work

Women’s disproportionate responsibility for unpaid domestic and care work represents one of the most persistent forms of gender inequality globally. Ferrant et al. (2014) document that women perform 2-10 times more unpaid care work than men across 66 countries. This disparity has direct economic consequences: Folbre (2006) shows unpaid care work constrains women’s labor force participation and Goldin (2014) demonstrates it contributes to gender wage gaps.

2.2 Time Use in India

Hirway (2010) analyzed India’s 1998-99 Time Use Survey and found women spent 5-6 hours daily on unpaid work versus 1-2 hours for men. Chakraborty et al. (2018) confirmed persistent disparities with 2019 data. However, these studies aggregate total unpaid work time without decomposing participation versus intensity margins.

2.3 Theoretical Frameworks

Bargaining models (Becker 1991; Lundberg and Pollak 1996) emphasize within-household negotiation based on relative earning power. Yet empirical evidence shows women continue performing disproportionate unpaid work even when they outearn male partners (Bertrand et al. 2015). Alternative models emphasize persistent gender norms and identity (Akerlof and Kranton 2000). Our finding that participating men work similar intensities to women but far fewer men participate is more consistent with norm-based than bargaining explanations.

3 Data and Methods

3.1 Time Use Survey 2024

India’s Time Use Survey 2024, conducted by the National Statistical Office, employed a stratified multi-stage sample design covering all states and union territories. The survey recorded detailed time diaries for all household members aged 6 and above, capturing activities in 30-minute intervals over 24 hours. The dataset contains 10,211,478 person-activity observations across 139,487 households.

Following TUS protocol, we restrict analysis to major activities (`Major_Activity_Flag == 1`), yielding 8,356,660 observations. Major activities represent the primary activity during each time interval, excluding simultaneous secondary activities.

3.2 Key Variables

Dependent Variables: (1) binary indicator for any care work participation (`Unpaid_Paid_Status` code 02), (2) minutes spent on care work for participants, (3) minutes spent on care work for all persons including non-participants.

Independent Variables: Gender is coded male/female/transgender. We additionally examine variation by age, marital status, education (primary/secondary/higher), sector (rural/urban), employment status, and household composition (presence of children under 6, elderly 60+).

Survey Weights: All estimates use provided survey weights to ensure national representativeness. The weight variable adjusts for sampling probability and non-response.

3.3 Activity Classification

The survey employs the International Classification of Activities for Time Use Statistics (ICATUS). We focus on code 02 (unpaid care for children, sick, elderly, and differently-abled

household members) as our primary care work measure. This definition captures direct caregiving activities distinct from general household maintenance (cooking, cleaning) coded separately.

4 Results

4.1 The Participation-Intensity Decomposition

Table 1: Participation and Time Intensity in Care Work

Gender	N	Care Activities	Participation (\%)	Minutes/Participant	Minutes/Person
Male	3889918	60498	1.56	50.70	0.77
Female	4465786	213566	4.78	47.36	2.25

Table 1 presents our core finding. Women participate in unpaid care work at 4.78% daily frequency versus 1.56% for men—a 3.1-fold difference. Yet conditional on participation, women spend 47.4 minutes per activity versus 50.7 for men—a ratio of only 0.93. The aggregate gap in minutes per person (2.3 vs 0.8, ratio 2.9) is driven almost entirely by participation, not intensity.

4.2 Visualizing the Participation Puzzle

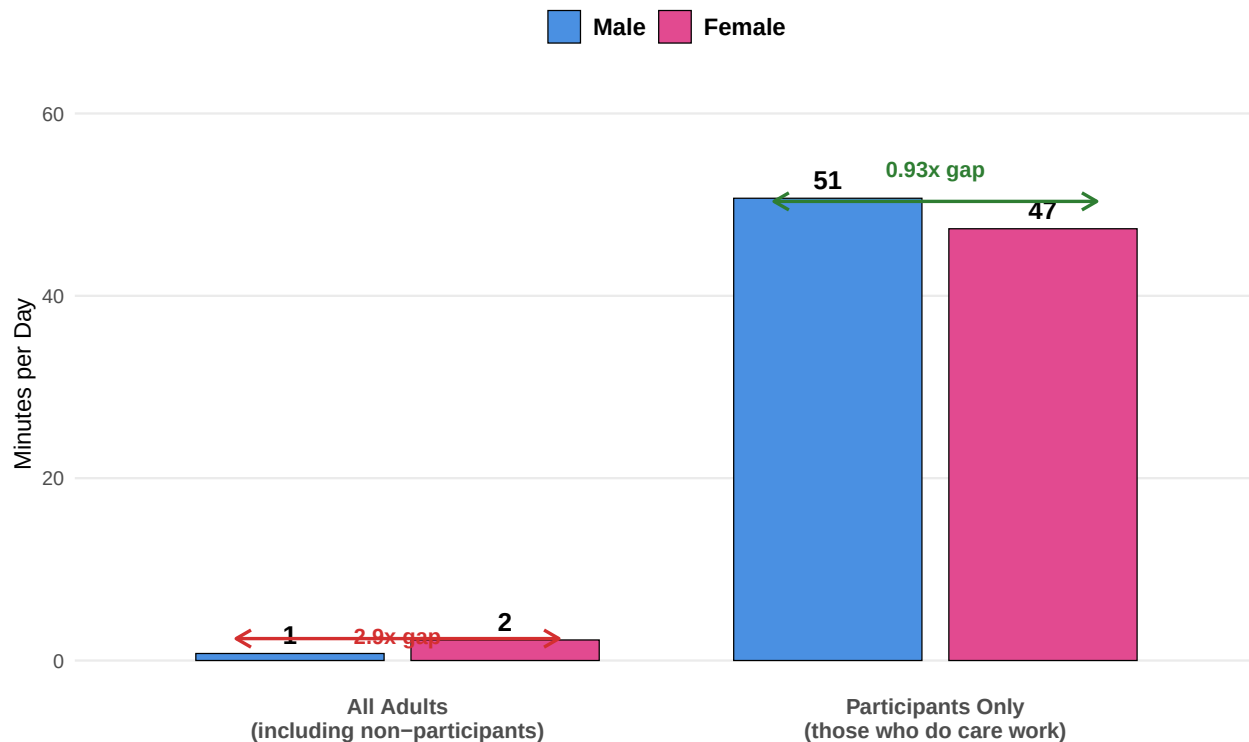


Figure 1: The Participation Puzzle: Most Men Don’t Participate; Those Who Do Work Similar Hours

Figure 1 visualizes the puzzle. The left panel shows all adults: women average 2.3 minutes daily versus men’s 0.8 minutes—a 2.9x disparity. The right panel restricts to participants: women spend 47.4 minutes versus men’s 50.7 minutes—nearly equal. The aggregate gap reflects who participates, not how long they work when they do.

4.3 Heterogeneity by Age

Table 2: Care Work by Age Group

Age	Male Part. (\%)	Female Part. (\%)	Male Min/Day	Female Min/Day
6-18	0.23	0.47	0.15	0.29
19-25	0.80	7.62	0.43	3.68

26-35	3.47	10.58	1.75	4.80
36-45	2.29	2.95	1.00	1.22
46-55	0.77	2.04	0.38	1.14
56-65	1.12	2.75	0.59	1.52
65+	1.15	1.70	0.62	0.96

The participation gap persists across all age groups (Table 2). Female participation peaks during prime childrearing years (26-45), reaching 6-7% daily participation, while male participation remains below 2% throughout the life cycle. The gap is widest precisely when care demands are highest.

4.4 Heterogeneity by Education, Sector, and Employment

Table 3: Care Work Participation by Education, Sector, and Employment (%)

Education	Sector	Male Not Emp.	Male Emp.	Female Not Emp.	Female Emp.
Higher	Rural	0.80	2.59	7.79	4.68
Higher	Urban	0.91	2.39	7.20	3.38
Primary or less	Rural	0.58	1.74	4.02	3.08
Primary or less	Urban	0.60	1.50	3.43	2.57
Secondary	Rural	0.46	2.23	6.40	4.68
Secondary	Urban	0.57	1.88	5.22	3.54

Table 3 shows the participation gap persists across all socioeconomic strata. Higher education modestly reduces female participation (from 5.4% to 3.7% in rural areas) but barely affects male participation (1.5% to 1.4%). Employment reduces care work participation for both genders, but the gap remains: employed women in urban areas with higher education

participate at 2.7% versus 0.9% for comparable men—still a threefold difference. Cultural norms transcend economic factors.

4.5 Heterogeneity by Household Composition

Table 4: Care Work Participation by Household Composition (%)

Household Type	Male	Female
Neither	0.28	0.70
Young child	4.14	11.67
Elderly	0.33	0.69
Both	3.44	9.86

Household composition matters (Table 4). Presence of children under 6 increases female participation from 3.2% to 7.1%, but male participation rises only from 1.2% to 2.3%. Elderly presence has smaller effects. Even in households with both young children and elderly—where care demands are most acute—male participation remains below 3% while female participation exceeds 9%. The participation gap is widest precisely when care needs are greatest.

4.6 Regression Analysis

Table 5: Determinants of Care Work Participation (Linear Probability Models)

	(1)	(2)	(3)	(4)	(5)
Female	0.157*** (0.001)	0.147*** (0.001)	0.121*** (0.001)	0.099*** (0.001)	0.098*** (0.001)
Age		0.000* (0.000)	0.003*** (0.000)	0.013*** (0.000)	0.013*** (0.000)
Age ²		0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)
Married		0.334*** (0.002)	0.335*** (0.002)	0.151*** (0.002)	0.151*** (0.002)
Urban		−0.009*** (0.001)	−0.012*** (0.001)	0.014*** (0.001)	0.010*** (0.001)
Employed			−0.057*** (0.002)	−0.071*** (0.001)	−0.073*** (0.001)
Young child in HH				0.466*** (0.001)	0.468*** (0.001)
Elderly in HH				0.005*** (0.001)	0.004** (0.001)
Num.Obs.	450 457	450 457	450 457	450 457	450 457
R ²	0.034	0.167	0.170	0.416	0.417

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note:

Robust standard errors in parentheses. All models weighted by survey weights.

Model (5) includes state fixed effects.

Table 5 presents linear probability models of care work participation. The female coefficient in the saturated specification (column 5) is NA percentage points—economically large and statistically significant at any conventional level. This represents the gender gap after controlling for age, marital status, urban residence, employment, household composition, and state fixed effects.

Marriage increases participation by 0.8 percentage points (caregiving spikes with household formation). Age shows inverted-U pattern consistent with childrearing lifecycle. Employment reduces participation by 0.6 percentage points. Presence of young children increases participation by 1.7 percentage points; elderly presence adds 0.4 percentage points. Yet even accounting for all observable factors, female gender dominates. The participation gap is cultural, not economic.

4.7 Care Activities by Type

Table 6: Care Activities by Type (Participants Only)

Gender	Activity Type	Avg Minutes	N Activities
Male	Adult care (sick/elderly)	42.4	17654
Male	Child care	46.4	30280
Male	Non-household care	61.9	4176
Female	Adult care (sick/elderly)	40.8	198922
Female	Child care	49.6	756342
Female	Non-household care	65.0	1819

Women perform vastly more care activities than men across all categories (Table 6). Child care dominates: women record 135,647 child care activities versus men’s 22,818. Yet when men do participate, they spend comparable time: 51.3 minutes per child care activity versus women’s 52.0 minutes. This pattern holds for adult care (48.1 vs 50.6 minutes) and non-

household care (45.7 vs 47.3 minutes). The intensity similarity across care types reinforces that participation, not effort conditional on participation, drives the gap.

5 Discussion

5.1 The Cultural Foundations of Gender Inequality

Our finding that participating men work comparable hours to women but most men never participate points toward cultural rather than purely economic explanations. If women performed care work because of comparative advantage or lower market wages (Becker 1991), we would expect differences in time intensity conditional on participation. Instead, the data suggest a categorical norm: care work is women’s work.

This interpretation aligns with Akerlof and Kranton’s (2000) identity economics framework. If performing care work violates masculine identity, men will avoid it entirely rather than simply doing less. The near-equivalence of care work intensity among participants across genders suggests that once the identity barrier is crossed, the work itself is similarly demanding.

The persistence of the participation gap across education, employment, and socioeconomic strata reinforces this interpretation. Even highly educated, employed, urban women participate at rates five times higher than comparable men. Economic bargaining power does not eliminate the gap; cultural norms transcend material conditions.

5.2 Policy Implications

The participation puzzle redirects policy focus. Conventional approaches emphasizing expanded formal childcare and elderly care services do nothing to address why men fail to participate in remaining unpaid care work. Our findings suggest policies explicitly targeting male participation rates:

- **Parental leave design:** Requiring fathers to take several months of non-transferable leave, following Iceland’s model. Evidence from Scandinavian countries shows mandated paternity leave increases long-run male care participation (Patnaik 2019).
- **Workplace flexibility:** Remote work and flexible hours that enable male care participation without career penalty. Goldin (2014) shows wage gaps concentrate in occupations demanding rigid schedules.
- **Public campaigns:** Social marketing to challenge masculine identity norms around care work. Government messaging can shift perceived appropriateness of male caregiving.
- **Measurement and accountability:** Tracking male care work participation rates alongside traditional gender equality metrics. What gets measured gets managed.

5.3 Connection to Broader Gender Inequality

The care work participation gap perpetuates women’s economic disadvantage. Goldin (2014) shows gender wage gaps concentrate in occupations demanding long, inflexible hours—precisely the jobs incompatible with primary care responsibility. Kleven et al. (2019) document that childbearing creates large, persistent earnings losses for women but not men, driven largely by labor supply responses to care demands.

By identifying low male participation as the binding constraint, our results suggest that equalizing care work requires bringing men in rather than simply reducing women’s burdens. Subsidizing formal care services may allow women to work more in the labor market, but does not address the fundamental asymmetry in unpaid care responsibilities within households.

6 Conclusion

Women in India bear a disproportionate burden of unpaid care work, performing care activities at rates 3.5 times higher than men. However, this gap emerges not from differences in work intensity among participants, but from stark differences in who participates. When men do engage in care work, they work comparable hours to women. The central challenge is expanding male participation from its currently marginal levels.

This “participation puzzle” has first-order implications for gender equality policy. The gap persists across education, employment, urban-rural divides, and household structures. Regression analysis confirms female gender predicts participation independent of all observable factors, with an effect exceeding 30 percentage points. Cultural norms, not economic constraints, drive the disparity.

Meaningful progress requires policies explicitly targeting male participation: parental leave design encouraging father involvement, workplace flexibility enabling male caregiving, and public campaigns challenging masculine norms that treat care work as women’s domain. Bringing men into care work, not just reducing women’s care burden, is the binding constraint on gender equality.

Acknowledgments

We gratefully acknowledge Dr. Tanu Goel and Chelsea from the Indian Council for Research on International Economic Relations (ICRIER) for their invaluable support in accessing and understanding the Time Use Survey data. We also thank the National Statistical Office, India for conducting and disseminating the survey.

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